

How Digital Workers See & Control Computers

Accessibility-Tree Snapshots: Semantic Understanding Over Brittle Visual Pixels

PAGE 28 / 42

The Semantic Revolution in Browser Automation

Traditional browser automation fails because web pages frequently change CSS classes, responsive layouts, and visual styling. Fathom bypasses visual fragility by operating browsers via the **Accessibility Tree** (the standard semantic model used by screen readers).

LEGACY PIXEL / CSS AUTOMATION (FRAGILE)

- **Obfuscated CSS:** Class names like `.sc-82f_btn` change on every frontend software deployment.
- **Pixel Drift:** Screen scaling or window resizing causes mouse clicks to hit empty whitespace.
- **High Latency:** Uploading high-res screenshots to vision models consumes massive bandwidth and tokens.

FATHOM SEMANTIC ACCESSIBILITY (ROCK SOLID)

- **Semantic Roles:** The agent sees functional elements: Button: "Login", TextBox: "Email".
- **Opaque Numerical Refs:** Direct addressing via stable tokens (e.g. @e14) regardless of visual layout.
- **10x Token Efficiency:** Lightweight accessibility snapshots use 90% fewer tokens than raw images.

Accessibility Snapshot Architecture

1. Chromium Page
Dynamic DOM & SPAs



2. ARIA Snapshot
Extracts semantic accessibility tree



3. Opaque Token Mapping
Assigns stable refs (@e1, @e2)



4. Agent Interaction
click(@e1), type(@e2, "text")

SCREEN-READER SIMPLICITY FOR AI MODELS

By translating complex visual layouts into clean, hierarchical text trees (role, name, value, state), foundation models can reason over web applications with 99%+ accuracy while consuming a fraction of the token budget required by vision models.

Layout-Agnostic Stability: Whether a website is redesigned, translated into Japanese, or resized to mobile viewport, the accessibility tree retains functional semantic integrity.